

8. Design of Block RAM IP using Xilinx Vivado

Aim

To design and configure a **Block Memory Generator IP core as a Single Port ROM** in Xilinx Vivado and simulate its operation to verify correct read access of stored data.

Software Required

- Xilinx Vivado Design Suite
- Vivado IP Catalog
- Vivado Simulator (XSIM)

Procedure

1. Create a New Project

- Open Xilinx Vivado.
- Create a new RTL project.
- Select the target FPGA device/board.
- Choose *Do not specify sources at this time*.

2. Create Block Design

- Open **IP Integrator** → *Create Block Design*.
- From the IP Catalog, add **Block Memory Generator (8.4)** IP.

3. Configure Block Memory Generator

- Double-click the IP and set:
 - **Interface Type:** Native
 - **Memory Type:** Single Port ROM
 - **ECC Type:** No ECC
 - **Algorithm:** Minimum Area
- Set required:
 - Memory depth
 - Data width
- (Optional) Load an **.coe or .mem file** to initialize ROM contents.

4. Connections

- Connect:
 - clka → system clock
 - addra → address input

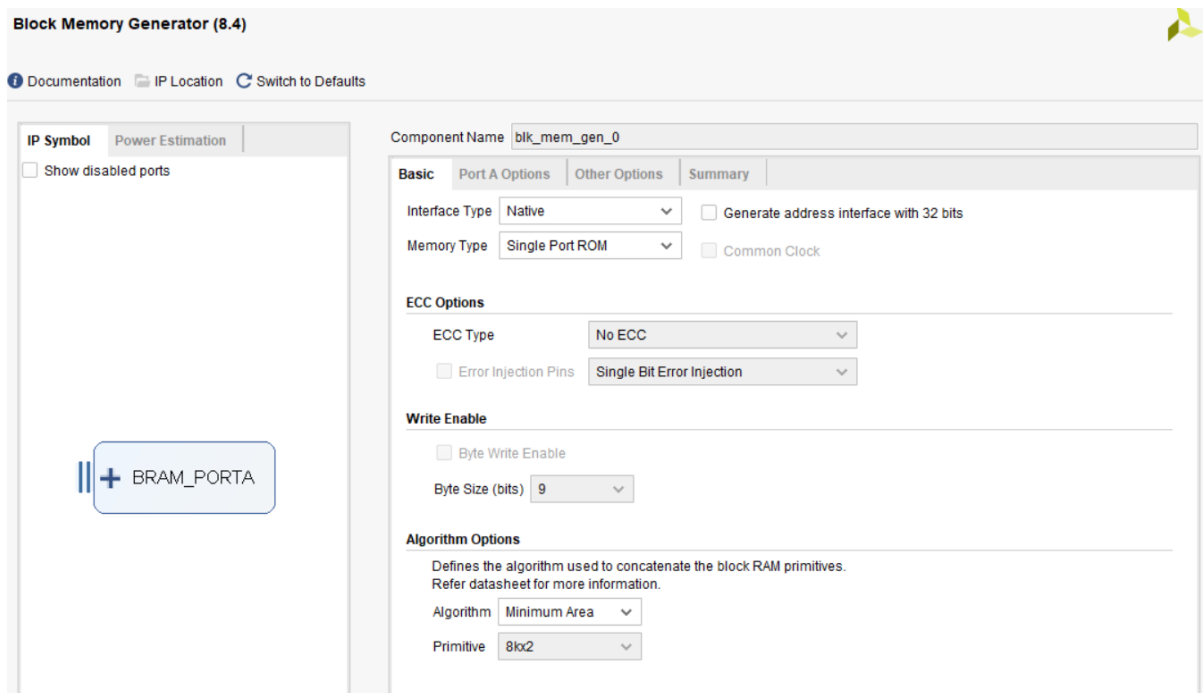
- douta → output port
- ena → logic '1'

5. Create HDL Wrapper

- Right-click the block design → *Create HDL Wrapper* → Let Vivado manage.

6. Simulation

- Write a simple testbench:
 - Generate clock.
 - Apply different addresses.
- Run **Behavioral Simulation**.
- Observe output data corresponding to each address.



Basic

Port A Options

Other Options

Summary

Memory Size

Port A Width

16

Range: 1 to 4608 (bits)

Port A Depth

16

Range: 2 to 1048576

The Width and Depth values are used for Read Operation in Port A

Operating Mode

Write First

Enable Port Type

Always Enabled

Port A Optional Output Registers

☐ Primitives Output Register
☐ Core Output Register

☐ SoftECC Input Register
☐ REGCEA Pin

Port A Output Reset Options

☐ RSTA Pin (set/reset pin)

Output Reset Value (Hex) 0

☐ Reset Memory Latch

Reset Priority CE (Latch or Register Enable)

READ Address Change A

☐ Read Address Change A

OK

Cancel

Re-customize IP

Block Memory Generator (8.4)

Documentation
IP Location
Switch to Defaults

IP Symbol

Power Estimation

Show disabled ports

+

BRAM_PORTA

Component Name blk_mem_gen_0

Basic

Port A Options

Other Options

Summary

Pipeline Stages within Mux 0 Mux Size: 1x1

Memory Initialization

☒ Load Init File

Coe File

i:/testing/fdp/fdpsrcs/sources_1/ip/coe_file.coe

Browse

Edit

☒ Fill Remaining Memory Locations

Remaining Memory Locations (Hex)

0

COE File Editor - coe_file.coe

Key	Value
memory_initialization_radix	10
memory_initialization_vector	101 102 103 104 105 106 107 108 109 110 111

Validate

Save

Save As...

Close

OK

Cancel

Testbench program

```
`timescale 1ns / 1ps

module tb_Block_RAM;

reg clk;

reg rst;

wire clk_out;

blk_mem_gen_0 uut (

    .clka(clk),

    .addra(4'd3),

    .douta()

);

always #5 clk = ~clk;

initial begin

    clk = 0;

    rst = 1;

    #20 rst = 0;

    #100;

    $stop;

end

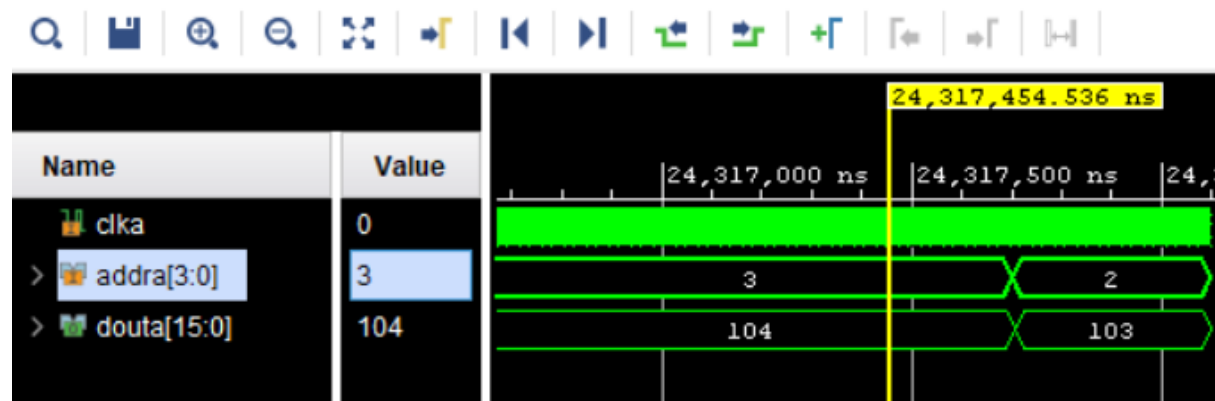
endmodule
```

Input

Radix: 10

Memory: 101 102 103 104 105 106 107 108 109 110 111

Testbench output



Result

The Block Memory Generator IP was successfully configured as a **Single Port ROM** and simulated in Vivado. For each applied address, the ROM produced the correct stored data at the output. The simulation waveforms confirmed proper read operation, validating the correct implementation of ROM using FPGA block memory. Thus, the design and verification of a ROM using the Block Memory Generator IP core were successfully completed.